

Algebra First-Year Exam, August 2015, Part 2

Answer four problems. (If you turn in more, the first four will be graded.) Within reason, you may quote theorems as long as you state them clearly.

1. Let R be a principal ideal domain. Prove that it satisfies the ascending chain condition on ideals.
2. This problem has three parts.
 - (a) Define Unique Factorization Domain.
 - (b) Define irreducible element of an integral domain R .
 - (c) Prove from your definitions, that if R is a Unique Factorization Domain and p is an irreducible element of R , then the ideal $P = (p)$ generated by p is a prime ideal of R .
3. This problem has two parts.
 - (a) Define Euclidean domain.
 - (b) Prove that every Euclidean domain is a principal ideal domain.
4. Let R be an integral domain and let M be an R -module. Suppose that N is an R -submodule of M . We denote by $\text{Ann}(M)$ the annihilator of M in R , so that $\text{Ann}(M)$ is an ideal of R .
 - (a) Prove that $\text{Ann}(M/N) \text{Ann}(N) \subseteq \text{Ann}(M)$.
 - (b) Give an example (with proof) where $\text{Ann}(M/N) \text{Ann}(N) \neq \text{Ann}(M)$.
 - (c) Give an example (with proof) where $\text{Ann}(M/N) \text{Ann}(N) = \text{Ann}(M)$.
5. How many distinct isomorphism types of abelian groups of order $2^4 \cdot 3^4 \cdot 5^4$ and exponent $2^3 \cdot 3 \cdot 5^2$ are there? Justify your answer carefully.
6. Let K be a splitting field for $x^4 + 2$ over $\mathbb{Q}$. Calculate the degree of $[K : \mathbb{Q}]$, justifying carefully each step.